



Efficacy of herbal tincture as treatment option for retained placenta in dairy cows

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ARTICLE INFO

Article history:

Received 4 May 2013

Received in revised form

22 December 2013

Accepted 24 December 2013

Available online 10 January 2014

Keywords:

Retained placenta

Traditional Chinese veterinary medicine

Tincture

Herbal therapy

Dairy cow

ABSTRACT

Retained placenta remains therapeutic challenge in cattle. Certain traditional medicines are believed to be able to alleviate retained placenta condition and improve overall fertility in cows. The aim of the present study was to evaluate the efficacy of an herbal tincture for treatment of retained placenta. The herbal tincture was extracted from a combination of *Herba Leonuri*, *Angelicae Sinensis Radix*, *Flos Carthami*, *Myrrha* and *Rhizoma Cyperi* by percolation with 70% ethanol to a concentration of 0.5 g crude herb/ml. Cows diagnosed with retained placenta ($n = 48$) were randomly divided into one of two treatment groups (A and B), with animals in group A ($n = 26$) receiving herbal tincture orally, and cows in group B ($n = 22$) receiving oxytetracycline infusion into the uterus. Eighty six cows with no clinically visible pathological conditions, given birth alone and with no retained placenta diagnosis were included into control group (C). Retained placenta was expelled within 72 h following initial treatment in 19 cows in group A, yet no cows in group B were recorded to expel placenta in the same time. The median number of days to first service (70.0 vs. 102.5 days; $P < 0.05$) and median number of days open (76.0 vs. 134.0 days; $P < 0.01$) were lower in group A than in group B. Percentage of cows pregnant within 100 days postpartum was the highest for animals in group A compared to controls (61.5% vs. 39.5%, $P < 0.05$), and for animals in group B (61.5% vs. 22.7%; $P < 0.01$). Herbal tincture used in the present study might facilitate expulsion of retained placenta and improve subsequent fertility, thus could present effective treatment option for retained placenta in cows.

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1. Introduction

Retained placenta is one of the most common puerperal disorders affecting the reproductive performance of dairy cows (LeBlanc, 2008). According to Paisley et al. (1986), Fourichon et al. (2000), and Drillich et al. (2003), retained placenta is defined as placenta that has not been expelled

within 12–24 h postpartum. The incidence of retained placenta varies from 3% to 18% overall (Paisley et al., 1986; Markusfeld, 1987; Eiler, 1997; Kimura et al., 2002; Han and Kim, 2005) and from 16% to 29.1% for cows in China (Cao and Qi, 2007; Guan and Gu, 2009). The negative effects of retained placenta include reduced milk yield, increased incidence of metritis and impaired subsequent fertility (Curtis et al., 1985; Holt et al., 1989; Laven and Peters, 1996; Stevens and Dinsmore, 1997). Therefore, effective treatments for retained placenta are crucial for improve puerperal healthy care in cows.

Many broad-spectrum antibiotic and hormonal therapies have been used to treat retained placenta in dairy cattle. However, the efficacy of these approaches

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is controversial, and some treatments might negatively affect subsequent reproductive performance (Stocker and Waelchli, 1993; Stevens and Dinsmore, 1997; Risco and Hernandez, 2003; Drillich et al., 2006, 2007; Goshen and Shpigiel, 2006). Consequently, alternative treatments have been considered. Collagenase therapy is one candidate approach (Eiler and Hopkins, 1993), although it is considered too costly for widespread use. In addition, Djuricic et al. (2012), and Zobel and Tkalčić (2013) reported that ozonated foam administered into the uterus is useful in treating retained placenta in cows resulting in beneficial effect on puerperal health and fertility. However, retained placenta may have several different etiologies, including abortion, dystocia, hypocalcemia, twin birth, induction of parturition, placentitis, nutritional disturbance and immunosuppression (Beagley et al., 2010); thus, retained placenta remains therapeutic challenge in cattle, and alternative options should be considered. One well-known advantage of traditional medicine in particular is that it provides holistic therapy for interrelated diseases (Jiang, 2005; Grayson, 2011).

According to traditional Chinese veterinary medicine (TCVM) theory, retained placenta falls within blood stasis syndrome category and Chinese herbal medicine is able to alleviate this condition in cows. A meta-analysis investigating the common characteristics of herbal formulas used to treat retained placenta showed that these formulas are typically associated with the therapeutic principle of “promoting blood circulation and removing blood stasis” (Yang et al., 2006). The herbal tincture used in the present study was designed according to this therapeutic principle and has been applied to treat retained placenta by experienced traditional veterinary medicine practitioners for decades. This herbal tincture consists of *Herba Leonuri*, *Angelicae Sinensis Radix*, *Flos Carthami*, *Myrrha* and *Rhizoma Cyperi* (Table A1 of Appendix A). According to TCVM theory, these herbs mixed as presented could resolve blood stasis syndrome caused by retained placenta. Previously described herbal tincture has been demonstrated not to be toxic if administered orally, according to Hodge and Sterner scale (CCOHS, 2005), in acute toxicity tests (maximum daily dose = 40 g/kg BW) in mice (Wang et al., 2011).

The aim of the present study was to evaluate the efficacy of the herbal tincture for the treatment of retained placenta and to assess subsequent fertility in cows.

2. Materials and methods

2.1. Herbal tincture preparation

The herbal tincture in this study was composed of five herbs – *Herba Leonuri*, 120 g; *Angelicae Sinensis Radix*, 30 g; *Flos Carthami*, 20 g; *Myrrha*, 25 g; and *Rhizoma Cyperi*, 30 g – that were administered as a single dose for a cow weighing 500 kg (equal to 0.45 g crude herb/kg BW). Herb quality criteria were congruent to the Veterinary Pharmacopoeia of the People's Republic of China (Chinese Veterinary Pharmacopoeia Commission, 2010). The herbs were pre-treated by washing, drying, chopping, pulverizing and screening with a 3-mm mesh. Following that, herbs were mixed as previously described. The mixture

was percolated with 70% ethanol at room temperature until colorless, and the extracted hydroalcoholic liquid were evaporated under reduced pressure at 40 °C. The condensed extract was adjusted with 60% ethanol to obtain an herbal tincture with a final concentration of 0.5 g crude herb/ml and 40% ethanol.

2.2. Animals

The trial was conducted from March 2011 to September 2012 in a commercial dairy farm of 1480 Holstein cows in Lanzhou city, China. Cows were housed in free-stall barns. At least 1 week before the expected calving, cows were moved into a free-stall barn with straw bedding. After parturition, cows were switched to a fresh lactation total mixed ration (TMR) over the course of 2 weeks. The TMR consisted of corn silage and hay as forage, with a corn gluten-cottonseed meal-based concentrate and minerals and vitamins and formulated to meet or exceed the nutritional requirements of early lactating dairy cows according to NRC (2001). Cows were fed TMR *ad libitum* during the study period. Cows were milked three times daily. The average milk yield varied between 5800 and 7012 kg (fat: $3.9 \pm 0.4\%$; protein: $3.2 \pm 0.3\%$).

2.3. Enrollment criteria

In the present study, retained placenta is defined as placenta that has not been expelled within 12 h postpartum. Treatment groups included cows given birth from March 2011 till March 2012, with retained placenta diagnosis. The enrolled cows were 2 to 7 years old (1–5 lactations) and diagnosed with dystocia, given twins or having pathogenic cold. For the purpose of presented study, dystocia diagnosis was established according to the protocol described by Schuenemann et al. (2011) with assistance score between 1 and 2. Pathogenic cold is defined as pathogenic factor in traditional Chinese medicine that results in a syndrome characterized by poor appetite, pricking pain in the lower abdominal region, bluish-purple lips and cold limbs (Editorial Committee of Encyclopedia of China's Agricultural, 1991). In the present study, pathogenic cold could lead to blood stasis syndrome, which might block the cotyledon–caruncle attachment, resulting in Qi (a term in traditional Chinese medicine referring to the energy flow or life force that is said to pervade all living things) and blood disorders that might interfere with the normal separation and release of the placenta (Yang et al., 2006; Lü et al., 1992). In addition, cows with no clinically visible pathological conditions, given birth alone and with no retained placenta diagnosis were included into control group (C). All enrolled cows had a body condition score between 2 and 4.5 at the time of calving according to the criteria of Edmonson et al. (1989).

2.4. Experimental protocol

Out of 620 cows given birth during the study period, retained placenta was diagnosed in 126. Eighty six cows having placenta expelled within 8 h following the birth and with no clinically visible puerperal disorders were included

into C group (controls). Fifty-three cows having retained foetal membranes diagnosis, dystocia, having foetotomy performed, given twins and diagnosed with pathogenic cold were included into two treatment groups (A, $n = 28$ and B, $n = 25$). During the study period two animals in group A, and three cows in B group were excluded due to the other reasons. Finally, three groups of animals (A, $n = 26$; B, $n = 22$; C, $n = 86$) were created for further statistical analysis.

During the experimental period, placental expulsion was evaluated at 2-h intervals by perineal inspection, and no attempts were made to manually remove placenta. Obstetrical examinations were performed at 72 h following the initial treatment. Prior to examination, vulva was washed using water and soap, and disinfected using an iodine compound followed by 70% alcohol. Examination gloves were lubricated and hand inserted into vagina, up to the external cervical orifice to confirm the uterine condition and placental expulsion. Cows in treatment groups, and not expelled placenta within 72 h following initial treatment were monitored for possible puerperal disorders until uterine involution occurred with lochia and placenta expelled as liquid material and debris.

Three experienced farm managers monitored cows for the clinically visible oestrous signs for the three times daily, starting from the day 25 until day 80 following the partus. Oestrus diagnosis was established in animals having oestrous mucus visible in vagina, and/or cervix with open cervical orifice using vaginoscopy, and after finding the dominant follicle larger than 20 mm in diameter using transrectal palpation and/or ultrasonography on one of the ovaries. Voluntary waiting period for the oestrus to occur was 80 days following the birth, and after that was induced using cloprostenol sodium (2 ml:0.2 mg, Suzhou SUMU animal medical industry co. Ltd) in amount of 4 ml intramuscularly. Pregnancy was confirmed by transrectal palpation and ultrasonography between days 40 and 48 following the last Artificial insemination (AI). AI was performed by experienced inseminators within 24–28 h after observed estrus, with semen deposited into the uterine horn ipsilateral to the ovary bearing the dominant follicle. Inseminations were performed throughout the year at the dairy farm in which the trial was conducted. Pregnancy was confirmed by transrectal palpation and ultrasonography between days 40 and 48 following the last AI. The number of days from partus until first estrus, number of AI's until pregnancy, and number of days open were recorded by the same inseminators for further statistical analysis.

2.5. Efficacy measurements

The primary outcome measure was the effect of the herbal tincture on the expulsion of retained placenta in cows. The condition was defined as cured if placenta was expelled within 72 h following initial treatment. Secondary outcome measures included days to first AI, days open, the first AI conception proportion, AI submission proportion, percentage of cows pregnant within 100 days postpartum and percentage of cows pregnant within 200 days postpartum.

2.6. Exclusion criteria

Animals suffering from diseases such as caesarean section during previous or current calving, mastitis (increased somatic cell counts $>200,000$ CFU/ml of milk) for more than 2 months post-calving, displaced abomasum, polyarthrititis, laminitis, as well as animals having extremely defective construction of hind limbs were excluded from the study due to the possible influence on results. In addition, animals developing fever ($>40^{\circ}\text{C}$) within 10 days post-calving, or any other pathological uterine condition that required parenteral antibiotic treatment were excluded from the study.

2.7. Treatment protocol

The treatment began after retained placenta diagnosis was established with herbal tincture given orally at dose of 450 ml (equal to 0.45 g crude herb/kg BW) to animals in group A. In cases with placenta not expelled within 4–6 h following the first dose, a second dose was given in the same amount. Cows of the group B received oxytetracycline infusion into the uterus (2000 mg in 250 ml of normal saline) once daily over 3-day period. No cows suffering from retained placenta were used as a control group for ethical reasons.

2.8. Statistical analysis

The data were analyzed using SPSS for Windows (version 17.0, IBM SPSS Statistics; USA). The first AI conception proportion, AI submission proportion, percentage of cows pregnant within 100 days postpartum and percentage of cows pregnant within 200 days postpartum were compared between groups using the chi-squared test. Normality of distribution of the median number of days to first AI and median number of days open were tested by the Shapiro–Wilk normality test. Differences among the groups in all variables were analyzed by the Kruskal–Wallis test and multiple comparisons of mean ranks. The experimental data are presented as the medians and interquartile ranges, and the level of significance was set at $\alpha = 0.05$.

3. Results

3.1. Effect of the herbal tincture on the release of retained placenta

The primary outcome measure was the time required for placenta being expelled regarding to the group (treatment). The placenta was expelled within 8 h following partus for all cows in group C. Out of the 26 cows in group A, 19 cows were recorded to expel placenta within 72 h following the beginning of the treatment. On the other hand, none of the cows in group B expelled placenta within the same time, and the retained placenta was ultimately expelled as liquid material and debris. In addition, three cows and two cows expelled their placenta on the 4th and 5th days, respectively, following initial treatment.

Table 1

Influence of the group on postpartal fertility criteria.

Therapy success criteria	Group A	Group B	Group C
No. of cows	26	22	86
AI submission proportion (%)	80.8	77.3	88.4
First AI conception proportion (%)	57.1	35.3	53.9
Median number of days to first service (interquartile range)	70.0 ^a (52.5–86.0)	102.5 ^b (65.8–134.5)	75.0 ^a (57.3–97.8)
Median number of days open (interquartile range)	76.0 ^A (66.5–100.0)	134.0 ^{B,b} (85.7–182.5)	93.0 ^a (66.5–130.5)
Percentage of cows pregnant within 100 days post partum	61.5 ^{A,a}	22.7 ^B	39.5 ^b
Percentage of cows pregnant within 200 days post partum	80.8	63.6	76.7

Values within the same row marked with different letters in superscript differ significantly: a, b = $P < 0.05$; A, B = $P < 0.01$.

No placental expulsion was recorded for the other four cows in group A, and the retained placenta was ultimately expelled as liquid material and debris.

3.2. Reproductive performance evaluation

The effects of the herbal tincture on reproductive performance parameters are shown in Table 1. The median number of days to first AI ($P < 0.05$) and median number of days open ($P < 0.01$) were lower for cows in group A compared to group B. Percentage of cows pregnant within 100 days postpartum was the highest for cows in group A compared to controls ($P < 0.05$), and for cows in group B ($P < 0.01$). The AI submission proportion was the highest for cows in group C, slightly lower for cows in group A, and the lowest in group B ($P > 0.05$). In addition, the median number of days to first AI, median number of days open, the first AI conception proportion and percentage of cows pregnant within 200 days postpartum were superior for cows in group A compared to controls ($P > 0.05$). The first AI conception proportion and percentage of cows pregnant within 200 days postpartum were higher for cows in group A compared to cows in group B ($P > 0.05$).

4. Discussion

According to the working hypothesis of the present study, oral administration of the herbal tincture effectively facilitated the spontaneous expulsion of retained placenta. Eiler (1997) reported that retained placenta does not resolve until average of 7 days in the absence of treatment. As the key component of the herbal tincture, *Herba Leonuri* can stimulate uterine contractions (Chan et al., 1983; Ma et al., 2000; Zhao et al., 2008) and modulate uterine hemorheology and microcirculation (Liu, 2008). It can be suggested that *Angelica sinensis* modulates host immunity through anti-inflammatory, immunomodulatory, anti-oxidative and antithrombotic properties (Yin et al., 1980; Zhao and Moghadasian, 2008; Chao and Lin, 2011; Yang et al., 2012). It is possible that herbal tincture treatment provided more favorable environment for an earlier placenta detachment through modulating host immunity, stimulating uterine contractions and improving the uterine condition. In the present study, 19 of 26 cows (73.08%) expelled placenta within 72 h following initial treatment with the herbal tincture. On the other hand, none of the animals receiving oxytetracycline infusion into the uterus (B group) expelled placenta within the same time. As previously reported by Drillich et al.

(2006, 2007), and Goshen and Shpigel (2006) that while antibiotics can be beneficial in treating endometritis, it is unlikely that they will result in an earlier release of fetal membranes, the results in the present study are in concordance to cited studies. Thus, the herbal tincture given orally can be considered as a treatment option for treatment of retained placenta to replace the antibiotic therapy in food-producing animals and eliminate the withdrawal time associated with antibiotic residues.

In the present study, oral administration of the herbal tincture could lower the number of days to first AI and the number of days open. The findings might indicate that more cows resumed cyclicity and exhibited uterine recovery sooner after calving. According to Ho et al. (2011) and Chang et al. (2013), herbal therapy is beneficial for the uterine recovery process following delivery. It is possible that herbal tincture treatment facilitated the uterine recovery process in cows by improvement in the uterine environment in consequence of an earlier placental detachment and a balanced uterine condition. In addition, the herbal tincture treatment elevated the proportion of first AI conception and the percentage of cows pregnant within 100 days postpartum. Furthermore, the percentage of cows pregnant within 200 days postpartum was highest for cows in herbal tincture group compared to controls. The pregnancy rate is a good overall measure of reproductive performance. Thus, the results demonstrated that the herbal tincture treatment improved the overall fertility in cows with retained placenta. This could be result of the synergistic effect of *Herba Leonuri*'s ability to improve uterus conditions (Chan et al., 1983; Ma et al., 2000; Zhao et al., 2008); and *Angelica Sinensis* Radix and *Flos Carthami* having immunomodulatory, anti-inflammatory and antithrombotic properties (Yin et al., 1980; Zhao and Moghadasian, 2008; Chao and Lin, 2011; Yang et al., 2012; Jun et al., 2011); *Myrrh* and *Rhizoma Cyperi* having analgesic and anti-inflammatory properties (Gupta et al., 1971; Dolara et al., 1996). In the present study, cows having retained placenta treated with the herbal tincture have similar or even superior reproductive performance compared to the control group (C) which indicates the effectiveness of herbal tincture therapy. Furthermore, the use of the herbal tincture to treat retained placenta in cows is interesting and warranted in a time where restrictions on the use of antimicrobial drugs are increasingly imposed on animal agriculture for lack of residues. From a clinical perspective, herbal tincture therapy can be considered as a treatment option for treatment of retained placenta. Despite its limited statistical power, the present study

Table A1Component herbs of the herbal tincture in this study^a.

Herb name	Traditional medicinal uses ^a	Modern pharmacology ^b	Bioactive phytochemical ^b
Yi Mu Cao (<i>Herba Leonuri</i>)	Used as a remedy for female reproductive disorders to treat delayed menstruation, menstrual pain and premenstrual tension	Emmenagogue, anti-spasmodic, cardiac tonic and anti-coagulant activities	Stachydrine
Dang Gui (<i>Angelicae Sinensis Radix</i>)	A key herb for maintaining female health and treating irregular menstruation, amenorrhea and dysmenorrhea	Smooth muscle contraction, cardiovascular and anti-thrombotic activities	Ligustilide, Ferulic acid
Hong Hua (<i>Flos Carthami</i>)	Used for amenorrhea, dysmenorrhea and wounds with pain and swelling	Anti-platelet, anti-inflammatory, analgesic and anti-pyretic activities	Hydroxysafflor yellow A
Mo Yao (<i>Myrrha</i>)	Treatment of menopausal symptoms, headache and indigestion and aiding tissue repair	Analgesic, anti-coagulant and cytoprotectant activities	Essential oil
Xiang Fu (<i>Rhizoma Cyperi</i>)	Used as a remedy for menstrual disorders, stomach ache and diarrhea	Anti-bacterial, anti-nociceptive and anti-inflammatory activities	Essential oil

^a A traditional Chinese herbal formula usually contains different quantities of several Chinese herbs. Individual herbs typically play different roles in a herbal formula and are termed the Monarch (Jun), Minister (Chen), Adjuvant (Zuo) and Guide (Shi). The 'Monarch' agent(s) is the key component of the herbal formula and are responsible for its major therapeutic effects against the principal symptom of a related disease. The mission of the 'Minister' agent(s) is to potentiate the effect of the 'Monarch' agent(s) or treat the accompanying symptoms, whereas the role of the 'Adjuvant' agent(s) is usually to enhance the therapeutic effects and mitigate the adverse effects of the 'Monarch' and/or 'Minister' agent(s). The task of the 'Guide' agent(s) is to harmonize the activities of all the other ingredients and/or to facilitate their uptake into a specific organ or target tissues.

In the herbal tincture, as the 'Monarch' agents, Yi Mu Cao (*Herba Leonuri*) activates the blood circulation to regulate the menses, and Dang Gui (*Angelicae Sinensis Radix*) vitalizes and invigorates the blood to regulate menstruation and alleviate pain. Hong Hua (*Flos Carthami*), as the 'Minister' agent, further serves to activate the blood to dispel blood stasis in and around the uterus. Mo Yao (*Myrrha*) serves as the 'Adjuvant' agent that dispels stagnated blood from abdominal masses. Xiang Fu (*Rhizoma Cyperi*) modulates Qi circulation and resolves stagnation to alleviate pain and also regulates and harmonizes the other herbs in the formula while providing mild pain relief. All the herbs work together, exhibiting a synergistic action of "promoting blood circulation and removing stasis" to resolve the blood stasis caused by retained placenta and promote placental expulsion.

^a An analysis and summary based on the data in the Compendium of Materia Medica.

^b Summary based on the data in the Pharmacology of Traditional Chinese Medicine (unless otherwise specified).

provided certain valuable information regarding herbal therapy for retained placenta in dairy cows.

5. Conclusions

The results of this study indicated that oral administration of this herbal tincture (0.45 g crude herb/kg BW) effectively facilitated the spontaneous expulsion of retained placenta and improved the reproductive performance of cows diagnosed with retained placenta at 12 h postpartum. This herbal treatment thus represents a potential effective alternative postpartum therapy for cows with retained placenta. Future investigations will focus on the broader involvement of the herbal tincture in health-related quality of life for postpartum cows and the mechanisms underlying its therapeutic effects.

Acknowledgment

This paper was supported by grants from the Dairy Industry Technology and Systems Projects (CARS-37-06), Special Fund for Agro-scientific Research in the Public Interest (201303040-01) and the Major Special Projects of Gansu Province (1102NKD0202). The authors thank Mr. Guanglu Shi and their staff for diligent animal care.

Appendix A. Appendix A

Table A1.

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